Vectors I Basics Worksheet: Solutions

Compiled by Maximus Pung Jun

Basics of vectors: **i**, **j**, **k** notation, magnitude, unit vectors, parallel vectors, collinear points, the ratio theorem, the scalar product and the vector product.

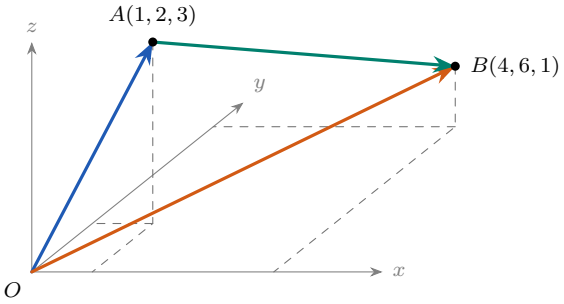
1 i, j, k Notation and Column Vectors

1 Write $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$ as a column vector, and write $\mathbf{b} = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}$ in terms of **i**, **j** and **k**.

$$\mathbf{a} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}$$

$$\begin{aligned} \mathbf{b} &= 4\mathbf{i} + 0\mathbf{j} - 1\mathbf{k} \\ &= 4\mathbf{i} - \mathbf{k} \end{aligned}$$

2 The points A and B have coordinates $(1, 2, 3)$ and $(4, 6, 1)$. Find $\overrightarrow{AB}$ in terms of **i**, **j** and **k**.



$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\ &= \begin{pmatrix} 4 \\ 6 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix} \\ &= 3\mathbf{i} + 4\mathbf{j} - 2\mathbf{k} \end{aligned}$$

3 **(Tricky)** Given $\mathbf{p} = 2\mathbf{i} + \mathbf{k}$ and $\mathbf{q} = -\mathbf{i} + 3\mathbf{j}$, find $\mathbf{r}$ such that $2\mathbf{p} - 3\mathbf{q} + \mathbf{r} = \mathbf{0}$. Give $\mathbf{r}$ in terms of **i**, **j** and **k**.

$$\mathbf{p} = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{q} = \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix}$$

$$\begin{aligned} \mathbf{r} &= 3\mathbf{q} - 2\mathbf{p} \\ &= 3\begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix} - 2\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} -3 \\ 9 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} -7 \\ 9 \\ -2 \end{pmatrix} \\ &= -7\mathbf{i} + 9\mathbf{j} - 2\mathbf{k} \end{aligned}$$

For the following statements, state whether it is true or false.

(a) $3\mathbf{i} + 2\mathbf{k} = \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$ **False**

$3\mathbf{i} + 2\mathbf{k} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} \neq \begin{pmatrix} 3 \\ 2 \\ 0 \end{pmatrix}$

(b) The $\mathbf{j}$-component of $4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ is -1 **True**

$4\mathbf{i} - \mathbf{j} + 2\mathbf{k} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}$

(c) The position vector of $P(1, -2, 0)$ is $\mathbf{i} - 2\mathbf{j}$ **True**

$\overrightarrow{OP} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} = \mathbf{i} - 2\mathbf{j}$

(d) $\overrightarrow{AB} = \overrightarrow{OA} - \overrightarrow{OB}$ **False**

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

(e) $\mathbf{i} + \mathbf{j} + \mathbf{k} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ **True**

$\mathbf{i} + \mathbf{j} + \mathbf{k} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

2 Magnitude

1 Find $|2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}|$.

$$\begin{aligned}|2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}| &= \sqrt{2^2 + (-3)^2 + 6^2} \\ &= \sqrt{49} \\ &= 7\end{aligned}$$

2 The points A and B have coordinates $(1, 2, -1)$ and $(3, -4, 2)$. Find the distance AB.

$$\begin{aligned}\overrightarrow{AB} &= \begin{pmatrix} 3 \\ -4 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix} \\ AB = \left|\overrightarrow{AB}\right| &= \sqrt{2^2 + (-6)^2 + 3^2} \\ &= \sqrt{49} \\ &= 7\end{aligned}$$

3 **(Tricky)** Given $\mathbf{a} = \begin{pmatrix} t \\ 2 \\ t-1 \end{pmatrix}$ and $|\mathbf{a}| = 3$, find the possible values of t.

$$\begin{aligned}|\mathbf{a}|^2 &= t^2 + 2^2 + (t-1)^2 \\ 9 &= t^2 + 4 + t^2 - 2t + 1 \\ 0 &= 2t^2 - 2t - 4 \\ 0 &= t^2 - t - 2 \\ 0 &= (t-2)(t+1) \\ t &= 2 \text{ or } t = -1\end{aligned}$$

For the following statements, state whether it is true or false.

(a) $|\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}| = 1 + 2 - 2 = 1$ **False**

$|\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}| = \sqrt{1^2 + 2^2 + (-2)^2} = 3$

(b) $|-\mathbf{a}| = |\mathbf{a}|$ for every vector $\mathbf{a}$ **True**

$|-\mathbf{a}| = |-1|\,|\mathbf{a}| = |\mathbf{a}|$

(c) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$ for all vectors $\mathbf{a}$, $\mathbf{b}$ **False**

$|\mathbf{i} + \mathbf{j}| = \sqrt{2} \neq |\mathbf{i}| + |\mathbf{j}| = 2$

(d) $|3\mathbf{a}| = 3|\mathbf{a}|$ **True**

$|3\mathbf{a}| = |3|\,|\mathbf{a}| = 3|\mathbf{a}|$

(e) $\left|\begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix}\right| = -5$ **False**

$\left|\begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix}\right| = \sqrt{0^2 + (-5)^2 + 0^2} = 5$

(f) $\left|\begin{pmatrix}1\\1\\1\end{pmatrix}\right| = 1$ **False**

$$\left|\begin{pmatrix}1\\1\\1\end{pmatrix}\right| = \sqrt{1^2+1^2+1^2} = \sqrt{3} \neq 1$$

(g) $|\mathbf{a}+\mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$ for all vectors $\mathbf{a}$, $\mathbf{b}$ **True**

$\overrightarrow{OA} = \mathbf{a},\ \overrightarrow{AB} = \mathbf{b},\ \overrightarrow{OB} = \mathbf{a}+\mathbf{b},\ OB \leq OA + AB\ (\triangle OAB)$

3 Unit Vectors

1 Find the unit vector in the direction of $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

$$\left|\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}\right| = \sqrt{2^2 + (-1)^2 + 2^2}$$
$$= 3$$
$$\text{unit vector} = \frac{1}{3}\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

2 Find the vector of magnitude 10 in the direction of $\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$.

$$\left|\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}\right| = \sqrt{3^2 + 0^2 + (-4)^2}$$
$$= 5$$
$$\text{vector} = 10 \cdot \frac{1}{5}\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$$
$$= \begin{pmatrix} 6 \\ 0 \\ -8 \end{pmatrix}$$

3 (Tricky) The points A and B have coordinates $(1, 0, 2)$ and $(3, -6, 5)$. Find the vector of magnitude 14 in the direction *opposite* to $\overrightarrow{AB}$.

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$
$$= \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$$
$$\left|\overrightarrow{AB}\right| = \sqrt{2^2 + (-6)^2 + 3^2}$$
$$= 7$$
$$\text{vector} = -14 \cdot \frac{1}{7}\begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$$
$$= \begin{pmatrix} -4 \\ 12 \\ -6 \end{pmatrix}$$

For the following statements, state whether it is true or false.

(a) $\begin{pmatrix}1\\1\\1\end{pmatrix}$ is a unit vector **False**

$$\left|\begin{pmatrix}1\\1\\1\end{pmatrix}\right| = \sqrt{3} \neq 1$$

(b) $\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$ is a unit vector **True**

$$\left|\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}\right| = \frac{1}{3}\sqrt{1+4+4} = 1$$

(c) $\mathbf{i}+\mathbf{j}$ is a unit vector **False**

$$|\mathbf{i}+\mathbf{j}| = \sqrt{1^2+1^2+0^2} = \sqrt{2} \neq 1$$

(d) $|\hat{\mathbf{a}}| = 1$ for every non-zero vector $\mathbf{a}$ **True**

$$|\hat{\mathbf{a}}| = \left|\frac{\mathbf{a}}{|\mathbf{a}|}\right| = \frac{|\mathbf{a}|}{|\mathbf{a}|} = 1$$

(e) $\mathbf{a}$ and $\hat{\mathbf{a}}$ are parallel **True**

$$\mathbf{a} = |\mathbf{a}|\,\hat{\mathbf{a}}$$

4 Parallel Vectors ($\mathbf{a} = \lambda\mathbf{b}$)

1 Show that $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6 \\ 3 \\ -9 \end{pmatrix}$ are parallel.

$$\begin{aligned} \mathbf{b} &= \begin{pmatrix} -6 \\ 3 \\ -9 \end{pmatrix} \\ &= -3 \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \\ &= -3\mathbf{a} \end{aligned}$$

$\therefore$ $\mathbf{a} \parallel \mathbf{b}$ (shown)

2 Find p and q such that $\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$.

$$\begin{aligned} \begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix} &= \lambda \begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix} \\ 4 &= -2\lambda \\ \lambda &= -2 \\ p &= 3\lambda \\ &= -6 \\ -2 &= \lambda q \\ -2 &= -2q \\ q &= 1 \end{aligned}$$

3 **(Tricky)** Find all vectors of magnitude 6 that are parallel to $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

$$\begin{aligned} |\mathbf{a}| &= \sqrt{1^2 + 2^2 + (-2)^2} \\ &= 3 \\ \text{vector} &= \pm 6 \cdot \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \\ &= \pm \begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix} \end{aligned}$$

For the following statements, state whether it is true or false.

(a) If $\mathbf{a} \parallel \mathbf{b}$, then $\mathbf{a} = \mathbf{b}$ **False**

$2\mathbf{i} \parallel \mathbf{i},\ 2\mathbf{i} \neq \mathbf{i}$

(b) $\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$ **True**

$$\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix} = -2 \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$$

(c) $\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$ **False**

$$\frac{2}{1} = \frac{4}{2} \neq \frac{6}{4}$$

(d) $\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix}$ is parallel to $\begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$ **True**

$$\begin{pmatrix} 3 \\ 0 \\ -6 \end{pmatrix} = -3 \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$

(e) If $\mathbf{a} = -2\mathbf{b}$, then $\mathbf{a}$ and $\mathbf{b}$ are not parallel since they point in opposite directions **False**

$\mathbf{a} = \lambda\mathbf{b},\ \lambda = -2$

5 Collinear Points

1 Show that the points $A(1,2,3)$, $B(3,5,4)$ and $C(7,11,6)$ are collinear.

$$\begin{aligned}\overrightarrow{AB} &= \begin{pmatrix}3\\5\\4\end{pmatrix} - \begin{pmatrix}1\\2\\3\end{pmatrix}\\ &= \begin{pmatrix}2\\3\\1\end{pmatrix}\\ \overrightarrow{BC} &= \begin{pmatrix}7\\11\\6\end{pmatrix} - \begin{pmatrix}3\\5\\4\end{pmatrix}\\ &= \begin{pmatrix}4\\6\\2\end{pmatrix}\\ &= 2\overrightarrow{AB}\end{aligned}$$

$\therefore\ \overrightarrow{AB} \parallel \overrightarrow{BC},\ B$ common

$\therefore\ A, B, C$ collinear (shown)

2 For the points in Question 1, find the ratio $AB : BC$.

$$\begin{aligned}\overrightarrow{BC} &= 2\overrightarrow{AB}\\ BC &= 2AB\\ AB : BC &= 1 : 2\end{aligned}$$

3 (Tricky) The points $P(2,-1,0)$, $Q(4,1,3)$ and $R(a,7,b)$ are collinear. Find a and b.

$$\begin{aligned}\overrightarrow{PR} &= \lambda\overrightarrow{PQ}\\ \begin{pmatrix}a-2\\8\\b\end{pmatrix} &= \lambda\begin{pmatrix}2\\2\\3\end{pmatrix}\\ 8 &= 2\lambda\\ \lambda &= 4\\ a-2 &= 2(4)\\ a &= 10\\ b &= 3(4)\\ &= 12\end{aligned}$$

For the following statements, state whether it is true or false.

(a) If $\overrightarrow{AB} \parallel \overrightarrow{CD}$, then A, B, C and D are collinear **False**

$A(0,0,0), B(1,0,0), C(0,1,0), D(1,1,0):\ \overrightarrow{AB} = \overrightarrow{CD}$

(b) If $\overrightarrow{AB} = 3\overrightarrow{AC}$, then A, B and C are collinear **True**

$\overrightarrow{AB} \parallel \overrightarrow{AC},\ A$ common

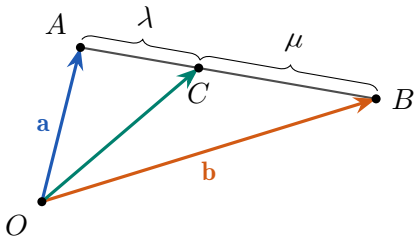
(c) $A(0,0,0)$, $B(1,1,1)$ and $C(2,2,3)$ are collinear **False**

$$\overrightarrow{AC} = \begin{pmatrix}2\\2\\3\end{pmatrix} \neq \lambda\begin{pmatrix}1\\1\\1\end{pmatrix}$$

(d) The points with position vectors $\mathbf{a}$, $2\mathbf{a}$ and $-\mathbf{a}$ are collinear ($\mathbf{a} \neq \mathbf{0}$) **True**

$2\mathbf{a} - \mathbf{a} = \mathbf{a},\ -\mathbf{a} - \mathbf{a} = -2(\mathbf{a})$

6 Ratio Theorem



In the True/False statements, C lies on AB with $AC : CB = \lambda : \mu$, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.

1 Let $\overrightarrow{OA} = \mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\overrightarrow{OB} = \mathbf{i}+2\mathbf{j}+3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

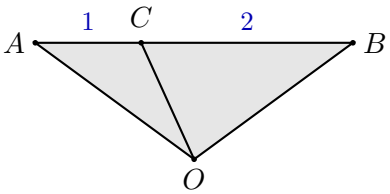
(a) $AC : CB = 1 : 2$,

(b) $3AC = 2CB$,

(c) $2AB = 5CB$,

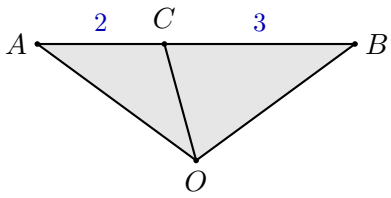
(d) Given instead that C lies on AB produced, and $3AC = 7BC$.

(a)



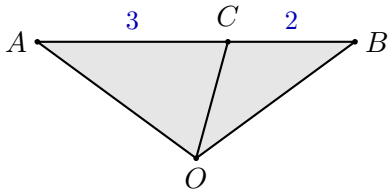
$$\overrightarrow{OC} = \frac{2\overrightarrow{OA} + \overrightarrow{OB}}{3}$$
$$= \begin{pmatrix} 1 \\ \frac{4}{3} \\ \frac{5}{3} \end{pmatrix}$$
$$\therefore\ C\left(1, \tfrac{4}{3}, \tfrac{5}{3}\right)$$

(b)



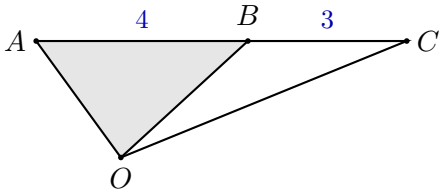
$$\frac{AC}{CB} = \frac{2}{3}$$
$$\overrightarrow{OC} = \frac{2\overrightarrow{OB} + 3\overrightarrow{OA}}{5}$$
$$= \begin{pmatrix} 1 \\ \frac{7}{5} \\ \frac{9}{5} \end{pmatrix}$$
$$\therefore\ C\left(1, \tfrac{7}{5}, \tfrac{9}{5}\right)$$

(c)



$$\frac{AB}{CB} = \frac{5}{2}$$
$$\overrightarrow{OC} = \frac{3\overrightarrow{OB} + 2\overrightarrow{OA}}{5}$$
$$= \begin{pmatrix} 1 \\ \frac{8}{5} \\ \frac{11}{5} \end{pmatrix}$$
$$\therefore\ C\left(1, \tfrac{8}{5}, \tfrac{11}{5}\right)$$

(d)



$$\frac{AC}{BC} = \frac{7}{3}$$
$$\overrightarrow{OB} = \frac{3\overrightarrow{OA} + 4\overrightarrow{OC}}{7}$$
$$\overrightarrow{OC} = \frac{7\overrightarrow{OB} - 3\overrightarrow{OA}}{4}$$
$$= \begin{pmatrix} 1 \\ \frac{11}{4} \\ \frac{9}{2} \end{pmatrix}$$
$$\therefore\ C\left(1, \tfrac{11}{4}, \tfrac{9}{2}\right)$$

2 Find the position vector of the midpoint M of AB, where A and B have coordinates $(3, -1, 5)$ and $(-1, 5, 1)$.

$$\overrightarrow{OM} = \frac{1}{2}\left[\begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix} + \begin{pmatrix} -1 \\ 5 \\ 1 \end{pmatrix}\right]$$
$$= \frac{1}{2}\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$
$$= \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

For the following statements, state whether it is true or false.

(a) $\overrightarrow{OC} = \dfrac{\mu\mathbf{a} + \lambda\mathbf{b}}{\lambda + \mu}$ **True**

$\overrightarrow{OC} = \mathbf{a} + \frac{\lambda}{\lambda+\mu}(\mathbf{b} - \mathbf{a}) = \frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$

(b) $\overrightarrow{OC} = \dfrac{\lambda\mathbf{a} + \mu\mathbf{b}}{\lambda + \mu}$ **False**

$\overrightarrow{OC} = \frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$

(c) If $\lambda : \mu = 1 : 1$, then $\overrightarrow{OC} = \dfrac{1}{2}(\mathbf{a} + \mathbf{b})$ **True**

$\overrightarrow{OC} = \frac{1\mathbf{a}+1\mathbf{b}}{2}$

(d) If $\lambda : \mu = 1 : 3$, then $\overrightarrow{OC} = \dfrac{3\mathbf{a} + \mathbf{b}}{4}$ **True**

$\overrightarrow{OC} = \frac{3\mathbf{a}+1\mathbf{b}}{1+3}$

(e) If $\lambda : \mu = 2 : 1$, then $\overrightarrow{OC} = \dfrac{2\mathbf{a} + \mathbf{b}}{3}$ **False**

$\overrightarrow{OC} = \frac{1\mathbf{a}+2\mathbf{b}}{3}$

7 Scalar Product

1 It is given that $\mathbf{a} = \begin{pmatrix}1\\2\\3\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}-3\\2\\7\end{pmatrix}$, $\mathbf{c} = \begin{pmatrix}4\\-5\\-1\end{pmatrix}$. Find the following

a) $\mathbf{a}\cdot\mathbf{b}$ c) $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})$ e) $\mathbf{a}\cdot(2\mathbf{c})$ g) $\mathbf{a}\cdot\mathbf{a}$
b) $\mathbf{b}\cdot\mathbf{a}$ d) $\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$ f) $(2\mathbf{a})\cdot\mathbf{c}$

Ans: a) 22 b) 22 c) 13 d) 13 e) -18 f) -18 g) 14

2

1) Points O, A, B are such that $\overrightarrow{OA} = \mathbf{i}+4\mathbf{j}-3\mathbf{k}$ and $\overrightarrow{OB} = 5\mathbf{i}-\mathbf{j}$
   (a) Find the size of angle AOB
   (b) Find the size of angle OAB.
   Ans: (a) 87.8° (b) 133.9°
2) It is given that $\mathbf{c} = \begin{pmatrix}2\\3\\6\end{pmatrix}$, $\mathbf{d} = \begin{pmatrix}1\\-2\\3\end{pmatrix}$ and $\mathbf{e} = \begin{pmatrix}3\\2\\-2\end{pmatrix}$. Find the angle between the vectors $\mathbf{c}$ and
   (a) the vector $\mathbf{d}$
   (b) the vector $\mathbf{e}$
   (c) the positive y-axis
   Ans: (a) 57.7° (b) 90° (c) 64.6°
3) It is given that two vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy the following relationship:
   $$|\mathbf{a}||\mathbf{b}| = 6.$$
   Find the value of $\mathbf{a}\cdot\mathbf{b}$ for each of the following scenarios.
   (a) The angle between $\mathbf{a}$ and $\mathbf{b}$ is 60°.
   (b) The angle between $\mathbf{a}$ and $\mathbf{b}$ is 120°.
   (c) Vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.
   Ans: (a) $\mathbf{a}\cdot\mathbf{b} = 3$ (b) $\mathbf{a}\cdot\mathbf{b} = -3$ (c) $\mathbf{a}\cdot\mathbf{b} = 0$

3

1) Referred to the origin O, the position vectors of the points A and B are $\mathbf{i}-\mathbf{j}+2\mathbf{k}$ and $2\mathbf{i}+4\mathbf{j}+\mathbf{k}$ respectively. Show that OA is perpendicular to OB.
2) The position vectors $\mathbf{a}$ and $\mathbf{b}$ are given by
   $$\mathbf{a} = 2p\mathbf{i}+3p\mathbf{j}+6p\mathbf{k} \quad \text{and} \quad \mathbf{b} = \mathbf{i}-2\mathbf{j}+2\mathbf{k}$$
   where $p > 0$. It is given that $|\mathbf{a}| = |\mathbf{b}|$.
   (i) Find the exact value of p.
   (ii) Show that the vectors $(\mathbf{a}+\mathbf{b})$ and $(\mathbf{a}-\mathbf{b})$ are perpendicular.
   Ans: (i) $p = \dfrac{3}{7}$

4

1) Given that $\mathbf{a} = 2\mathbf{i}+\mathbf{j}+2\mathbf{k}$ and $\mathbf{b} = -\mathbf{i}+2\mathbf{k}-\mathbf{k}$, find the length of projection and the projection vector of
   (a) $\mathbf{a}$ onto $\mathbf{b}$,
   (b) $\mathbf{b}$ onto $\mathbf{a}$.
2) With reference to the origin O, the points A, B and C have position vectors given by
   $$\mathbf{a} = \mathbf{i}+2\mathbf{j}-\mathbf{k},\ \mathbf{b} = 3\mathbf{i}+\mathbf{k} \text{ and } \mathbf{c} = 2\mathbf{i}+2\mathbf{j}+3\mathbf{k}$$
   respectively. Find the length of projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.

Ans:

1a) $\frac{2}{\sqrt{6}}, \frac{1}{3}\begin{pmatrix}-1\\2\\-1\end{pmatrix}$ 1b) $\frac{2}{3}, \frac{2}{9}\begin{pmatrix}2\\1\\2\end{pmatrix}$

2) $\frac{2}{3}$

For the following statements, state whether it is true or false.

(a) (Cancellation law) $\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c} \implies \mathbf{b}=\mathbf{c}$ **False**

$\mathbf{a}=\mathbf{i},\ \mathbf{b}=\mathbf{j},\ \mathbf{c}=\mathbf{k}:\ \mathbf{a}\cdot\mathbf{b}=0=\mathbf{a}\cdot\mathbf{c},\ \mathbf{b}\neq\mathbf{c}$

(b) $\mathbf{a}\cdot\mathbf{b}=\mathbf{c}\cdot\mathbf{d} \implies \mathbf{a}=\dfrac{\mathbf{c}\cdot\mathbf{d}}{\mathbf{b}}$ **False**

$\mathbf{a}=\mathbf{b}=\mathbf{i},\ \mathbf{c}=\mathbf{d}=\mathbf{j}:\ \mathbf{a}\cdot\mathbf{b}=1=\mathbf{c}\cdot\mathbf{d},\ \dfrac{\mathbf{c}\cdot\mathbf{d}}{\mathbf{b}}=\dfrac{1}{\mathbf{i}}$ undefined

(c) $\mathbf{a}\cdot\mathbf{b}=0 \implies \mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$ **False**

$\mathbf{a}=\mathbf{i},\ \mathbf{b}=\mathbf{j}:\ \mathbf{a}\cdot\mathbf{b}=0,\ \mathbf{a}\neq\mathbf{0},\ \mathbf{b}\neq\mathbf{0}$

8 Vector Product

1

1) Find $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix}$

2) Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, find

(a) $\mathbf{a} \times \mathbf{b}$; (c) $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$; (e) $\mathbf{a} \times (2\mathbf{b})$; (g) $\mathbf{a} \times \mathbf{a}$;
(b) $\mathbf{b} \times \mathbf{a}$; (d) $\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$; (f) $(2\mathbf{a}) \times \mathbf{b}$;

Ans:
1) $\begin{pmatrix} -29 \\ 17 \\ 26 \end{pmatrix}$
2) a) $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ b) $\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ c) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$ d) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$ e) $\begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$ f) $\begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$ g) $\mathbf{0}$

2

1. Three non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy $\mathbf{a} \times 3\mathbf{b} = 2\mathbf{a} \times \mathbf{c}$. Show that $\mathbf{a}$ and $3\mathbf{b} - 2\mathbf{c}$ are parallel.
2. Find the scalar product and vector product of the following vectors, stating whether they are parallel, perpendicular, or neither.
   (i) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$
   (ii) $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + \mathbf{k}$
   (iii) $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ -8 \\ 4 \end{pmatrix}$

Ans: 2) Neither; Perpendicular; Parallel

3 Using the Vector Product, find the acute angles between the following vectors:

a) $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$ b) $\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$ c) $\begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$

Ans: a) 51.0° b) 46.4° c) 87.0°

4

1) Relative to an origin O, two point A and B have position vectors given by $\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$. Find the following:
   a) The shortest distance from A to OB
   b) The shortest distance from B to OA
2) Relative to the origin O, two points A and B have position vectors given by $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ respectively.
   (i) Find the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.
   (ii) Hence, or otherwise, find the shortest distance from B to OA.

3) Relative to the origin O, the points A, B and C have position vectors given by $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = \mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{c} = 3\mathbf{i} - \mathbf{j} + \mathbf{k}$. Find the shortest distance from C to the line joining A and B.
4) With respect to an origin O, the points A and B have position vectors $\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ and $5\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ respectively. F is the foot of perpendicular of point B on the line that passes through A and is parallel to the vector $\mathbf{d} = 2\mathbf{i} + \mathbf{j} + 4\mathbf{k}$
   (i) Find the exact length of AF
   (ii) Find the exact length of BF
   (iii) Find the vectors $\overrightarrow{AF}$ and $\overrightarrow{FB}$.

Ans:

1) (a) 4.09 units (b) 4.51 units
2) (i) $2\sqrt{3}$ units (ii) $\sqrt{2}$ units
3) 2 units
4) (i) $\dfrac{11}{\sqrt{21}}$ (ii) $\dfrac{\sqrt{5397}}{21}$ (iii) $\overrightarrow{AF} = \begin{pmatrix} \frac{22}{21} \\ \frac{11}{21} \\ \frac{44}{21} \end{pmatrix}$, $\overrightarrow{FA} = \begin{pmatrix} \frac{62}{21} \\ -\frac{32}{21} \\ -\frac{23}{21} \end{pmatrix}$

5

1) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} -2 \\ 3 \\ 2 \end{pmatrix}$.
   Find the area of triangle OAB.
2) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, $\overrightarrow{OB} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$.
   Find the area of triangle OAB.
3) Relative to an origin O, the position vectors of the vertices A, B and C of a triangle are
   $$\mathbf{i} + \mathbf{j} - \mathbf{k},\ 2\mathbf{i} + \mathbf{k} \text{ and } -\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$
   respectively. By considering the displacement vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$, find
   (a) the exact area of triangle ABC, and
   (b) the exact length of the vector component of $\overrightarrow{AB}$ perpendicular to $\overrightarrow{AC}$.
4) The points A and B have coordinates $(5, 0, 5)$ and $(10, -5, 15)$ respectively. The point D is such that $OADB$ is a parallelogram and the point C divides AD in the ratio $2 : 3$. Find the exact areas of
   (i) the parallelogram $OADB$, and
   (ii) the quadrilateral $OCDB$.

Ans: 1) $\dfrac{\sqrt{154}}{2}$ units^2 2) $\dfrac{3\sqrt{2}}{2}$ units^2 3) (a) $\dfrac{\sqrt{21}}{2}$ units^2 (b) $\dfrac{\sqrt{105}}{5}$ units^2 4) (i) $25\sqrt{3}$ units^2 (ii) $20\sqrt{3}$ units^2

END OF WORKSHEET